

Module 02: Hardware Identifiers — MAC Addresses & Local Footprints

Many beginners confuse MAC addresses with IP addresses and misunderstand where hardware identifiers travel. Let's break down the mechanics at Layer 2 vs Layer 3.

■ 1. What is a MAC Address?

A **Media Access Control (MAC)** address is a 48-bit (6-byte) physical identifier permanently assigned to a Network Interface Card (NIC) by the manufacturer.

ORGANIZATION IDENTIFIER

00 : 1A : 2B

OUI (First 3 Bytes / 24 Bits)

Registered to Vendor (Intel, Realtek, Apple)

DEVICE SERIAL

3C : 4D : 5E

NIC Specific ID (Last 3 Bytes / 24 Bits)

Unique Physical Hardware Serial

■ 2. Where Does a MAC Address Travel? (The Layer 2 vs Layer 3 Rule)

2.1 The Hop-by-Hop Principle

- **MAC addresses operate exclusively at OSI Layer 2 (Data Link Layer).**
- **MAC addresses NEVER leave your local broadcast domain.**

HOP 1: YOUR PC → LOCAL ROUTER

SRC MAC: **00:11:22:33:44:55 (Your NIC)**

DST MAC: AA:BB:CC:DD:EE:FF (Router LAN)

STATUS: **Your MAC Visible ONLY on Local LAN**

ROUTER STRIPS ETHERNET FRAME

HOP 2: ROUTER → REMOTE WEBSERVER

SRC MAC: Router WAN Interface MAC

DST MAC: ISP Gateway Interface MAC

STATUS: **Your PC's MAC is 100% GONE**

- **Conclusion:** Remote servers, websites, and game hosts **cannot see your physical MAC address from an IP packet header**.
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■ 3. How Hardware Identifiers ARE Tracked & Leaked

If MAC addresses don't leave the local network in packet headers, how do adversaries and corporations use them to track you?

3.1 Local Wi-Fi Access Points & Captive Portals

- Public Wi-Fi (airports, hotels, cafes) logs your physical MAC address the instant you associate with the Access Point (AP).
- Even if you connect to a VPN immediately after authenticating, the network operator has already tied your device's physical hardware MAC, device hostname, and connection timestamps to your physical presence.

3.2 Client-Side Software & Telemetry (Layer 7 Exfiltration)

Websites running in standard web browsers cannot directly read your MAC address due to JavaScript sandbox restrictions. However:

- **Installed Applications (Steam, Discord, Game Anti-Cheats, EDRs, Malware):** Run with native OS privileges.
- They call native APIs (e.g., Windows WinAPI `GetAdaptersAddresses()` or `GetAdaptersInfo()`) to read your real MAC address, disk serial numbers, BIOS UUID, and CPU IDs.
- They package these identifiers into encrypted HTTPS requests (Layer 7 payload) and send them to their servers, creating an unshakeable hardware profile (HWID).

3.3 ARP Sniffing on Shared Networks

On any unsegmented LAN (e.g., open Starbucks Wi-Fi or dorm networks):

- Any machine running Wireshark or `arp-scan` can discover every active host's MAC address and private IP.
 - An attacker can execute **ARP Cache Poisoning (Man-in-the-Middle)** to redirect all your traffic through their machine before it reaches the router.
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■ 4. MAC Spoofing & Countermeasures

4.1 MAC Randomization

Modern mobile OSes (iOS, Android) and modern Windows/Linux desktop distributions feature built-in MAC randomization for Wi-Fi scanning and per-SSID connections.

4.2 Manual Spoofing

- **Linux:**

```
sudo ip link set dev eth0 down
sudo macchanger -r eth0
sudo ip link set dev eth0 up
```

- **Windows:**

- Device Manager -> Network Adapter -> Properties -> Advanced -> `Locally Administered Address` (or `Network Address`) -> Set to a custom 12-digit hex value.
- Note: The second character must be `2`, `6`, `A`, or `E` to indicate a locally administered unicast address (e.g., `02:XX:XX:XX:XX:XX`).

